How Much Is an Agent's History Worth?

A Law of Evaluation Transfer Across Agent Versions,
Confirmed in Six Pre-Registered Studies

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November 2026

Abstract

Every new version of an AI agent—a new model, prompt, or scaffold—is evaluated as if nothing had been learned before, although the outcomes of every earlier agent on the same tasks are already known. We organise this history as a *graph of agents*: nodes are agent versions, lineage edges link a version to its predecessor, and every node carries its outcome on each task of a shared suite. We ask how many tasks that history is worth, and answer with a closed-form law: testing m tasks of a new version with the help of the graph is as precise as testing $m/(1-\rho^2)$ tasks at random, where ρ^2 is the share of the new version's task-level variance already explained by the graph—φ^2 with the predecessor (double sampling), η^2 with the task difficulty seen by all earlier agents (stratified sampling). We turn the law into a three-move method: the graph *chooses* the tasks, a cross-fitted model-assisted estimator *estimates* the new version's success rate with calibrated intervals, and the same tasks *decide* whether to keep or replace the current version. Every hypothesis was pre-registered and frozen by a git tag before the data were examined. After a pilot on model versions, five confirmations cover 219 real version transitions never analysed before: agents on AppWorld, SWE-bench (Verified, Lite, test, multilingual), Multi-SWE-bench in seven languages, model versions on LiveCodeBench, and office agents on TheAgentCompany. The law predicts the observed efficiency gain within 3.9% (median) across all 219 transitions, and within 10% for 84% of them. With 20 graph-chosen tasks, the cross-fitted estimator matches about 44 random tasks, never lost to random testing in the last two confirmations (worst case $\times 1.08$ and $\times 1.45$), and its 90% intervals cover the truth 0.944 of the time; keep-or-replace decisions are wrong 32–54% less often than with random testing. Of 29 pre-registered hypotheses, 25 were supported; we report the 4 failures in full. We release a small kernel that implements the graph and reproduces these numbers on 89 real transitions.

1 Introduction

Agents change constantly. A coding agent ships a new model in March, a new scaffold in May, a new prompt every week. Each change raises the same question—is the new version better than the one in production?—and the standard answer is to re-run the full evaluation suite. On agentic benchmarks this is expensive: a single SWE-bench Verified run executes 500 repository-level tasks, each in its own container [8], and office-automation suites run long multi-step tasks in simulated companies [24].

Yet a team that evaluates a new version is never starting from nothing. It knows how the previous version did on every task, and how every other agent it ever tested did on the same tasks. This paper starts from a simple picture of that knowledge, inspired by the idea of an LLM-maintained wiki of interlinked pages [10]: a *graph of agents* in which each node is an agent version, lineage edges connect successive versions, and each node carries its task-level results. The question we study is quantitative:

When a new node arrives, how many tasks is the rest of the graph worth?

Contributions. This paper makes five contributions.

1. **A law of evaluation transfer** (Section 3). We show that classical results of survey sampling—double sampling [4, 16] and stratification—give a closed-form prediction of the efficiency gain an agent's history provides, $1/(1-\rho^2)$, with two measurable forms of ρ^2: the predecessor's φ^2 and the graph's η^2. The law turns "how much history is transferable" into a number that can be computed before deciding how many tasks to run.
2. **A three-move method** (Section 4). The graph chooses the tasks (stratification on the difficulty seen by earlier agents); a cross-fitted generalised regression estimator [19, 26] with effective-sample-size intervals [11] estimates the new version's rate; the same tasks decide keep-or-replace and flag whether the decision is sure.
3. **A pre-registered evidence chain** (Section 5). Six studies—one pilot and five confirmations on data never analysed before—each frozen by a git tag and SHA-256 hashes before the first statistic, with every development step done on previously seen data only and every deviation logged.
4. **Results on 219 real version transitions** (Section 6), including an atlas of what survives a version change and 4 pre-registered failures reported in full.
5. **A kernel** (Section 7) implementing the graph of agents—an append-only event log, a registry of slots and versions, and the operations `record`, `plan`, `estimate`, `decide`, `lint`—validated by replaying 89 real transitions.

2 Setting

A *task suite* is a fixed set of N tasks with binary outcomes (resolved or not). An *agent version* v has an outcome vector $y_v \in \{0,1\}^N$ and a success rate $p_v = \frac{1}{N}\sum_j y_{vj}$. The *graph* $\mathcal{G}$ at time t contains every version released before t whose outcome vector is fully known, and a lineage edge $v \to v'$ when v' is the next version of the same product (the same agent *slot*).

When a new version v' arrives, its outcome vector is unknown. We test it on $m \ll N$ *probe* tasks and must (i) estimate $p_{v'}$ with an interval, and (ii) decide whether $p_{v'} \geq p_v$. The baseline is *random testing*: m tasks drawn uniformly, estimated by their success rate (we use a Jeffreys-smoothed mean with Clopper–Pearson-type intervals). We measure every method by its *gain*:

$$\text{Gain} = \frac{\text{MSE(random testing with } m \text{ tasks)}}{\text{MSE(method with } m \text{ tasks)}}, \tag{1}$$

so that m tasks with the method are worth Gain $\times\, m$ random tasks. A gain below 1 is a loss.

3 A law of evaluation transfer

The predecessor. The predecessor v has been measured on all N tasks; the new version on m. This is exactly Neyman's double sampling [16]: a cheap auxiliary variable (here the predecessor's outcome) observed on a large first-phase sample, the expensive variable on a subsample. Let φ be the correlation between y_v and $y_{v'}$ over tasks (the phi coefficient of their 2×2 flip table).

Proposition 1 (Predecessor law)**.** *With the predecessor known on N tasks and m probes drawn at random, the double-sampling regression estimator has, to first order, variance $S^2(1-\varphi^2)/m + \varphi^2 S^2/N$ [4, ch. 12], where S^2 is the variance of $y_{v'}$ and tasks are viewed as draws from a population of tasks. Its gain over random testing is therefore*

$$\text{Gain}_\varphi = \frac{1}{1-\varphi^2+\varphi^2\, m/N}. \tag{2}$$

The gain depends on one number, φ^2: how much of the new version's task-level behaviour its predecessor already explains. A minor model update keeps φ^2 high; a rewrite of the agent's orchestration can erase it.

The graph. The rest of the graph carries a different kind of information: which tasks are hard. Sort the N tasks by the difficulty seen by all earlier agents, cut them into m contiguous strata of size $N_h \approx N/m$, and test one task per stratum. The success rate of the probes is unbiased for $p_{v'}$ because the strata are built from the history only.

Proposition 2 (Graph law)**.** *Under this stratified design, the gain over random testing is*

$$\mathrm{Gain}_\eta = \frac{(1 - m/N)\, S^2/m}{\frac{1}{m^2}\sum_h (1 - 1/N_h)\, S_h^2} \;\approx\; \frac{1}{1 - \eta_G^2}, \tag{3}$$

where S_h^2 is the variance of $y_{v'}$ within stratum h and η_G^2 is the share of its variance explained by the strata [4, ch. 5].

Both forms are instances of the same statement: *the history is worth $1/(1-\rho^2)$ times the tasks*, where ρ^2 is the fraction of the new version's task-level variance that the graph already explains. The individual results are classical; what is new is the identification of an agent's version history with the first phase of a two-phase design, the use of the graph of all earlier agents as the stratifier, and the empirical test that these closed forms predict the actual gains of agent evaluation (Section 6.1).

4 Method: choose, estimate, decide

Choose. Compute, for each task, the logit of the smoothed success rate of all earlier agents in the graph. Sort, form m strata, draw one task per stratum (Proposition 2). The selection uses the history only, so it is ignorable for model-based inference and keeps the probe mean unbiased.

Graph predictions. We fit a Bayesian logistic regression of the new version's probe outcomes on two graph features—the ecosystem difficulty and the predecessor's outcome—with priors $N(0, 2^2)$ on the intercept and $N(0, 1)$ on the slopes, averaging over the four covariate subsets with weights proportional to their marginal likelihoods (estimated by importance sampling with a multivariate-t proposal). This yields a probability $\hat{\pi}_j$ for every task. The model-based estimate of $p_{v'}$ adds the observed probe outcomes to the predicted outcomes of the $N - m$ untested tasks (finite-population prediction).

Estimate: cross-fitted GREG. Model-based predictions are efficient but can be biased when the model is wrong, which happens for versions that solve almost nothing. The generalised regression (GREG) estimator [19] corrects the predictions with the residuals observed on the probes and is design-unbiased whatever the model. Measuring residuals on the probes used to fit the model makes them too small and the intervals too narrow; we therefore cross-fit [26]: strata are split into two interleaved folds, the model fitted on one fold predicts the residuals of the other, and

$$\hat{p}_{v'} = \frac{1}{N}\sum_{j=1}^{N} \bar{\pi}_j \;+\; \frac{1}{N}\sum_{i \in \text{probes}} N_{h(i)} \left(y_{v'i} - \hat{\pi}_i^{(-k(i))}\right), \tag{4}$$

where $\bar{\pi}$ averages the two fold models and $\hat{\pi}^{(-k)}$ is the model not fitted on fold k. Its variance is estimated by pairing adjacent strata [4, §5A.12], floored at a quarter of the Jeffreys binomial variance, and converted into an interval for a proportion at the effective sample size, with a degrees-of-freedom correction and Jeffreys quantiles [11]; following Brown et al. [3], the lower bound is 0 when no probe succeeds and the upper bound 1 when all do.

Table 1: Pre-registered studies. "Transitions" are real version changes; "Tasks" is the task suite size.

Study	Source	Transitions	Tasks	Nodes are
v1 (pilot)	Sheng & Lu [21]	30	164–5,000	model versions
v2	AppWorld [22]	40	168 / 417	14 agents (4 scaffolds, 4 models)
v3	SWE-bench Verified [8]	89	500	175 public agents
v4	SWE-bench Lite	28	300	84 public agents
v5	Multi-SWE-bench [25] (7 lang.), SWE-bench test and multilingual	33	128–1,587	public agents
v6	LiveCodeBench [7], TheAgentCompany [24]	29	713 / 175	models; office agents
Total (confirmations v2–v6)		219		

Decide. Keep v or switch to v' by the sign of $\hat{p}_{v'} - p_v$, using the model-based estimate on the same graph-chosen probes; the decision is flagged as *sure* when the GREG interval excludes p_v.

Detect regressions sequentially. Probes arrive in batches of 10, up to 80; an alert is raised when the upper 95% bound of the predecessor-based (double-sampling) estimate falls below p_v, and testing stops early when the lower bound exceeds $p_v - 0.08$.

Predecessor-only baseline. To isolate what the graph adds, we also report a Bayesian double-sampling estimator that uses only the predecessor: it averages a model in which the new version ignores the predecessor and a model with separate success rates after predecessor successes and failures, weighted by Beta-binomial marginal likelihoods.

5 Protocol

Pre-registration. Each study was written as a protocol (hypotheses, thresholds, estimators, seeds, data rules) and an analysis script, committed together and frozen by a git tag (`prereg-v1` to `prereg-v6`) and SHA-256 hashes *before any statistic was computed on its data* [17]. Only format audits (file keys, task identifiers, submission names and dates) were allowed before the freeze. Every method change was developed on data already analysed in an earlier study, never on the data of the next confirmation. All deviations—including one accidental display of a single submission's total during a format audit—are logged in the repository. Hashes were re-verified after each run.

Data. Table 1 lists the sources. Transitions are defined mechanically from public submission names and dates: within a product lineage, each submission is paired with the latest strictly earlier submission of the same lineage; the graph of a new version contains only submissions released strictly before it. Tasks of SWE-bench test that also belong to Verified or Lite (analysed earlier) were removed before the freeze.

Replays. For every transition we simulate the evaluation of the new version 400 times (1,000 in the pilot): draw the probes (uniformly, or with the graph's stratified design, independently), apply every estimator to the same probes, and compare with the true rate measured on all N tasks. The primary budget is $m = 20$; $m = 40$ is reported as secondary. Coverage is the fraction of replays whose 90% interval contains the truth.

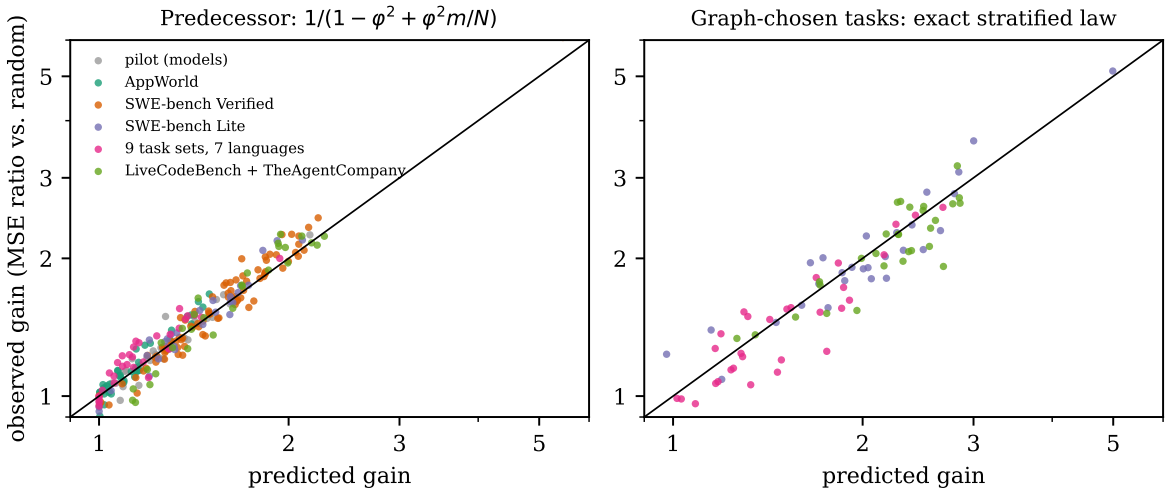


Figure 1: Predicted versus observed gain over random testing, $m = 20$. Left: predecessor law (2), all six studies (249 transitions). Right: graph law (3) for graph-chosen tasks, studies v4–v6 (90 transitions). The diagonal is perfect prediction.

6 Results

6.1 The law predicts the gain

Figure 1 plots predicted against observed gains. The predecessor law (2), evaluated at the true φ^2, predicts the gain of the double-sampling estimator within 3.9% (median absolute relative error) over the 219 confirmation transitions; the correlation of log-gains is 0.975. The error is 2.7%, 3.6%, 3.9%, 5.8% and 6.4% in the five confirmations, and 2.6% in the pilot: six pre-registered tests of the same law, all within the 20% threshold. The graph law (3) predicts the gain of graph-chosen tasks within 7.7% over 90 transitions (8.8%, 7.7%, 7.2% in v4–v6); the median $\eta^2_{\mathcal{G}}$ is 0.54 on SWE-bench Lite and 0.57 on LiveCodeBench, i.e. the history of other agents explains about half of a new version's task-level variance.

6.2 The graph is worth more than the predecessor

Figure 2 summarises the gains. Using only the predecessor gives median gains of $\times 1.12$ (AppWorld), $\times 1.53$ (Verified), $\times 1.47$ (Lite), $\times 1.22$ (v5) and $\times 1.56$ (v6). Using the whole graph raises them to $\times 1.39$, $\times 2.01$, $\times 1.95$, $\times 1.12$ and $\times 1.77$ with random tasks; on SWE-bench Verified the difference is significant ($p = 7.8 \times 10^{-14}$, paired Wilcoxon). Letting the graph also choose the tasks gives $\times 2.14$ on Lite and $\times 2.25$ on v6, where 20 graph-chosen tasks are worth about 43 and 44 random ones.

What survives a version change. The AppWorld design crosses four scaffolds with four models, so the type of change can be isolated (Table 2). A model upgrade at fixed scaffold (GPT-4 Turbo $\rightarrow$ GPT-4o) keeps a median $\varphi^2 = 0.301$; a scaffold change at fixed model keeps 0.081; a change of model family keeps 0.045. *Rewriting an agent's orchestration erases almost as much transferable information as changing model provider.* When the predecessor carries nothing, the rest of the graph still does: for family changes the graph gains $\times 1.49$ where the predecessor gains $\times 1.06$.

The lineage edge is real but subsumed. On SWE-bench Verified, a version shares more information with its predecessor than with an unrelated earlier agent of the same success rate

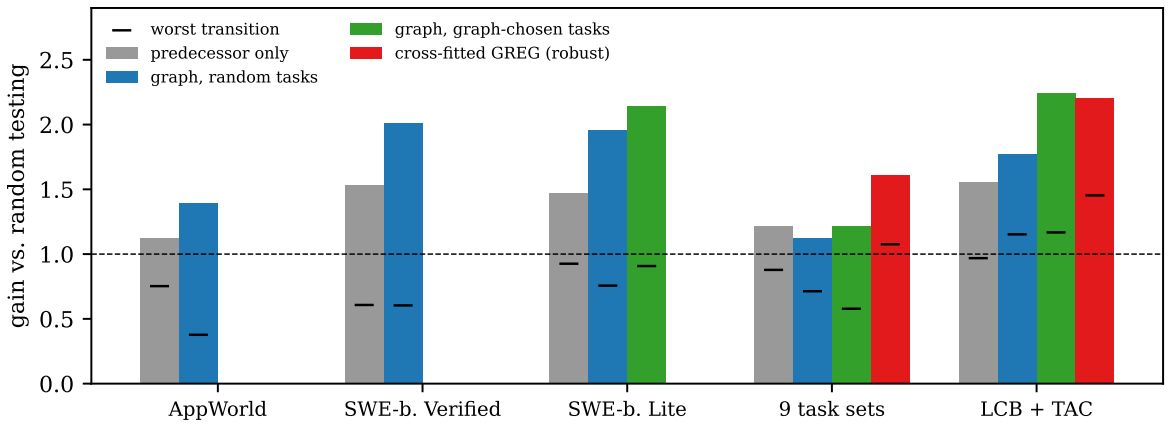


Figure 2: Median gain over random testing ($m = 20$) by study and method; black ticks mark the worst transition. The robust estimator was introduced in v5 (its non-cross-fitted predecessor in v4 is reported in the text).

Table 2: Atlas of transferable information on AppWorld (v2, pre-registered, Kruskal–Wallis $p = 0.0026$).

Type of change	n	median φ^2	gain, predecessor	gain, graph
Model upgrade, same scaffold	8	0.301	×1.47	×1.51
Scaffold change, same model	20	0.081	×1.10	×1.20
Model family change	12	0.045	×1.06	×1.49

(median φ^2 0.367 versus 0.316, pre-registered, $p < 10^{-6}$). But once the difficulty seen by the whole graph is used, adding the predecessor changes the median gain from ×2.006 to ×2.014: statistically detectable ($p = 0.038$) and practically negligible. The value lies in the graph, not in the lineage alone.

6.3 Robust and calibrated estimation

Model-based estimates can fail on versions that solve fewer than 10% of tasks: on SWE-bench Verified the graph estimator's worst transition was ×0.60 (a submission published with zero resolved tasks), while on the 85 transitions between 10% and 90% its worst case was ×1.25. Study v5 was dominated by this regime (21 of 33 new versions below 10%). There the cross-fitted GREG estimator gained ×1.61 (worst ×1.08; ×1.51 below 10%) with mean coverage 0.950 (minimum 0.812), whereas the model-based estimator's worst case was ×0.58. In v6, where the roles were fixed in advance, the cross-fitted estimator gained ×2.20 (worst ×1.45) with coverage 0.944 (minimum 0.895). The non-cross-fitted GREG of v4 never lost either (worst ×1.06) but its intervals under-covered (minimum 0.625), which motivated cross-fitting.

6.4 Decisions and regression detection

On transitions where the true rates differ by at least 5 points, keep-or-replace decisions from 20 graph-chosen tasks were wrong in 6.5%, 3.9% and 6.0% of replays (v4, v5, v6), against 14.1%, 7.6% and 8.8% with random testing (all pre-registered, $p < 0.05$). In real version histories, the sequential rule detected the 11 SWE-bench Verified versions that regressed by at least 10 points in 92% of replays with 30 probes on average, with 0.8% false alerts on the 73 transitions without a drop.

6.5 What failed

Four of the 29 pre-registered hypotheses were not supported (Appendix A). (i) In the pilot, predecessor-based estimates did not reduce decision regret significantly and (ii) sequential detection lacked power at 40 probes; the detection rule was retuned on development data and later confirmed in v2 and v3. (iii) On SWE-bench Verified the graph estimator lost badly on near-zero versions, so the hypothesis of no catastrophic loss failed. (iv) In v5, the model-based estimator with graph-chosen tasks did not stay above random testing on 90% of transitions, because most new versions solved almost nothing. Two further observations qualify our method. An unbiased estimator is not automatically a good decision rule: in v5, GREG's decisions were wrong in 24.0% of replays against 13.0% for random testing (9.3% versus 7.6% on differences of at least 5 points). The gap comes from transitions where both versions solve almost nothing: in the extreme case (both below 1%), GREG chose wrongly in 87% of replays and random testing in 0%, because the random baseline's prior pushes such estimates upward and happened to point the right way. And in v6 the pre-registered decider was not the best one: the plain mean of the graph-chosen tasks decided slightly better (4.1% versus 6.0%). The robust gain in decisions comes from *which tasks are tested*; which estimator should decide remains open.

7 The kernel: a graph of agents

The method needs a place where the graph lives. We implement it as a small kernel that observes and does not orchestrate. State is an append-only log of events (slot and version registrations with release dates and lineage, task results with their role and origin, probe plans); five operations act on it: `record` (one result or a whole evaluation), `plan` (the graph chooses m tasks for a new version and stores the plan, which fixes the stratum weights), `estimate` (cross-fitted GREG on the planned probes), `decide` (keep or replace, flagged sure or not), and `lint` (lineage inconsistencies, probes outside the plan, training tasks used as evidence). Each slot gets a Markdown page listing its versions, estimates and intervals.

Validation. Criteria fixed before the run: replaying the 89 SWE-bench Verified transitions through the kernel alone must reproduce the research gain within 10%, keep coverage above 0.88, and make fewer decision errors than random testing. The kernel obtained a median gain of $\times 2.18$, coverage 0.943, and decision errors of 6.0% versus 11.9% for random testing, with no lint finding.

Case study. On the nine real OpenHands versions in SWE-bench Verified, the kernel tested 180 tasks instead of 4,500. Its intervals contained the true rate for 100% of versions and 78% of its decisions were right; both wrong decisions had been flagged as not sure by the kernel itself, and every decision it flagged as sure was right.

8 Related work

Efficient evaluation. Item response theory [18] underlies tinyBenchmarks [13] and anchor points [23], which predict benchmark scores of language models from few items; active testing chooses test points adaptively [12]. For agents, Agent Psychometrics predicts task-level success on coding benchmarks by separating model and scaffold abilities [6]; Ndzomga [15] keeps tasks with mid-range historical pass rates; She and Lin [20] study an evolving production agent and deploy difficulty-stratified fixed subsets with IRT-based estimates; Johnson and Helm [9] estimate new agents' scores from cached traces of reference systems. These works report point accuracy (errors, rank correlations). We add a closed-form law that predicts the gain before

testing, calibrated intervals that survive model failure, the keep-or-replace decision, and pre-registered evidence.

Prediction-powered inference. PPI [1], PPI++ [2], cross-PPI [26] and stratified PPI [5] combine model predictions with a few labels for valid inference; Mani et al. [14] show that the benefit is not free in finite samples. Our estimator is the survey-sampling counterpart (GREG with cross-fitting under a stratified design), where predictions come from the graph of earlier agents rather than from a single predictor.

Version updates. Sheng and Lu [21] show that no single signal predicts which answers regress after an LLM update. We do not predict individual flips; we measure how much the history reduces the cost of estimating and deciding, which is what a team shipping a new version needs.

9 Limitations

The law assumes probes drawn at random (or within strata) from a fixed task suite; it says nothing about tasks outside the suite. Gains depend on the graph: with fewer than 20 earlier agents the median gain on SWE-bench Verified was markedly lower. Versions that solve almost nothing remain hard for decisions. Several sources share agents or products (SWE-bench splits; 21 of the 33 transitions in v5 are the same model change in different scaffolds and languages), and tasks within a repository are correlated. Release dates in v6 were recorded from memory by the research assistant (Acknowledgements). LiveCodeBench evaluates models rather than agents, and TheAgentCompany contributes only two transitions.

10 Conclusion

An agent's history is worth a computable number of tasks. Across 219 real version transitions in five pre-registered confirmations, the law $1/(1-\rho^2)$ predicted that number within a few percent, and a method in which the graph of earlier agents chooses the tasks, estimates with calibrated intervals and decides, roughly halved the cost of evaluating a new version. Evaluation does not have to start from zero at every release: the graph of agents remembers.

Code and data availability

Code, protocols, frozen analysis scripts, hashes, raw replay results and the kernel are in the accompanying repository; every number in this paper is generated from the result files by `paper/make_numbers.py`. The benchmark outputs are public: the AppWorld leaderboard bundles, the GitHub repositories `SWE-bench/experiments`, `multi-swe-bench/experiments`, `LiveCodeBench/submissions` and `TheAgentCompany/experiments`, and the data of Sheng and Lu [21].

Acknowledgements

The author designed the study, the agent-graph framing and the research programme, and takes full responsibility for the content. Claude (Anthropic), an AI assistant, was used extensively to write code, run the experiments and draft the text under the author's direction; it is not an author.

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A Ledger of pre-registered hypotheses

B Development history

Between studies, methods were changed only on data already analysed: the Bayesian model average of the predecessor-only estimator (synthetic worlds, before v1); the graph estimator and the sequential rule (pilot data, before v2); finite-population prediction, model averaging and an identifiability guard (AppWorld and pilot data, before v3); the stratified design and GREG (SWE-bench Verified and AppWorld, before v4); cross-fitting, the Korn–Graubard interval and the Jeffreys boundary convention (Verified, Lite, AppWorld, before v5). Variants tried and abandoned (an empirical-Bayes slope prior; fallback rules on the number of successes) are listed in the pre-registration documents.

Study	Hypothesis	Criterion (abridged)	Verdict
v1	Variability gate	φ^2 spread ≥ 0.20	supported
v1	H1 law	median error ≤ 0.20	supported (2.6%)
v1	H2 φ^2 depends on pair and benchmark	Holm-corrected tests	supported
v1	H3 lower decision regret	paired test	**not supported**
v1	H4 regression detection	power ≥ 0.80	**not supported**
v2	H1 law (agents)	median error ≤ 0.20	supported (2.7%)
v2	H5 graph > predecessor	Wilcoxon, $\geq$ 90% non-loss, coverage	supported (at the 90% limit)
v2	H2 φ^2 depends on type of change	Kruskal–Wallis	supported
v2	H4 sequential detection	power, probes, false alerts	supported (86%, 52 probes)
v3	H1 law	median error ≤ 0.20	supported (3.6%)
v3	H5 graph > predecessor	as v2	supported
v3	H6 no catastrophic loss	worst gain and coverage $\geq$ 0.80	**not supported**
v3	H7 predecessor adds to graph	Wilcoxon	supported, negligible effect
v3	H2 lineage edge carries information	Wilcoxon vs matched control	supported
v3	H4 detection of real regressions	power, false alerts	supported
v4	H1 law	median error ≤ 0.20	supported (3.9%)
v4	H8 graph law	median error ≤ 0.20	supported (8.8%)
v4	H9 graph-chosen tasks help	Wilcoxon, non-loss, coverage	supported
v4	H10 better decisions	Wilcoxon	supported
v4	H11 GREG no catastrophic loss	worst gain ≥ 0.80	supported
v5	H1 law	median error ≤ 0.20	supported (5.8%)
v5	H8 graph law	median error ≤ 0.20	supported (7.7%)
v5	H9 graph-chosen tasks help	Wilcoxon, non-loss, coverage	**not supported**
v5	H10 better decisions	Wilcoxon	supported
v5	H12 robust and calibrated	worst gain, coverage	supported
v6	U1 estimate: robust and calibrated	worst gain, coverage	supported
v6	U2 estimate: beats predecessor-only	Wilcoxon	supported
v6	U3 decide: beats random testing	Wilcoxon	supported
v6	U4 both forms of the law	median errors ≤ 0.20	supported